

An Investigation into Non-Small Cell Lung Cancer Final Report

Biomedical Engineering 1 W24

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Abstract

Non-small cell Lung Cancer (NSCLC) presents as a heterogeneous disease requiring personalized diagnostic and therapeutic strategies. Despite advancements in Targeted Therapies and Immunotherapies, predicting patient outcomes remains challenging. Biomarker testing plays a crucial role in customizing treatments to the specific characteristics of NSCLC patients by examining genes, proteins, and other substances. This personalized approach enhances treatment precision, mitigates ineffective treatments, and minimizes potential side effects, ultimately improving patient outcomes. Clinicians can tailor treatment strategies based on tumor response, mitigating resistance and offering NSCLC patients a clearer path to progression-free survival. Additionally, emerging techniques such as the efficacy of amivantamab in treating EGFR-mutated NSCLC, in combination with lazertinib, present promising solutions for addressing lung cancer mutations.

Background

Lung cancer is the leading type of cancer worldwide with cigarettes the main cause. Signs and symptoms don't typically occur at their earliest stages but occur when the disease is advanced and reaches metastasis (cancer spreading to other parts of the body). Risk factors include smoking, secondhand smoking, family history, harmful chemicals and asbestos such as nickel, arsenic, chromium exposure, air pollution like radon, family history of lung cancer (typically due to secondhand smoking), and previous radiation therapy. As the cancer spreads it can often spread to lymph blood vessels as well.

As further discussed in the critical need, another alternative is using biomarker testing, to have personalized testing in tailoring treatments against certain carcinomas. Furthermore, there are two types of lung cancer, small cell carcinoma and non-small cell carcinoma. Small cell carcinoma, appears flat and smaller than cancer cells in non-small cell lung

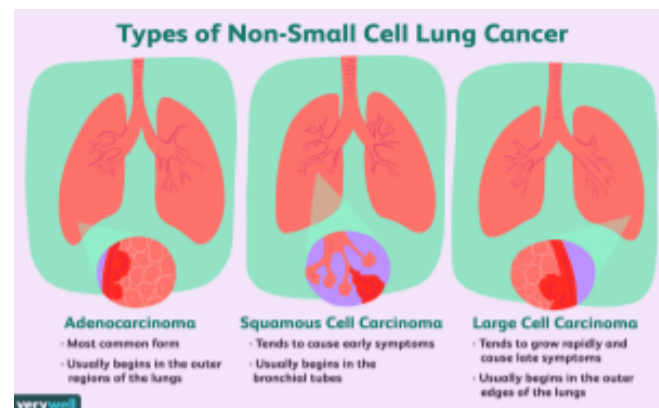


Fig 1- VeryWell Health

cancer, and typically grows faster and spreads to the lymph nodes. For the basis of this paper, since non-small carcinoma lung cancer is much more common, that is what will be discussed.

Within non-small carcinoma, there are three types, Adenocarcinoma (40% of cases) Squamous cell carcinoma (25% of cases), and large cell carcinoma (10% of cases) (Kenfield SA, Wei EK, Stampfer MJ, Rosner BA, Colditz GA (2008)). For adenocarcinoma, it starts in areas that normally generate mucus. In squamous cell lung cancer, it occurs in the central part of the lung and airway. This cancer is often more prevalent in patients who smoke.

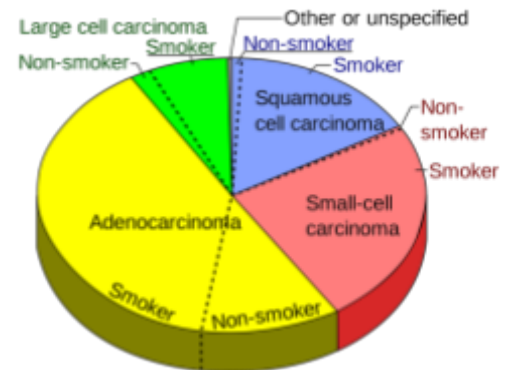


Fig 2- Distribution of cases of types of lung cancers

Diagnosis

Lung cancer diagnosis has gotten much more accurate and quicker over the last decades due to the vast amount of research that has been done. Here are suggested examples of the progress in diagnosing NSCLC.

Imaging Tests are where several scans can be used to look into the body without cutting it open, providing an idea of what is going on in the patient's body to the doctor. X-rays are useful in the detection of odd lumps or tumors in the lungs. Computerized tomography (CT) scans can give the doctor a clear and detailed picture of the lungs, showing the size as well as how far the tumor has spread. Magnetic resonance imaging (MRI) scans are also able to give a perspective of the spread of cancer.

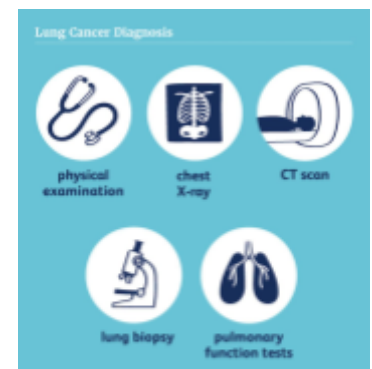


Fig.3- methods used for diagnosis of NSCLC

Needle Biopsy is essentially when a very narrow needle is inserted into the lung to remove tissue. Bronchoscopy uses a camera along with a thin tube to go into the body from the mouth or nose. Surgical biopsy is utilized to extract a bigger piece of tissue if it is required.

Furthermore, laboratory tests such as Sputum Cytology are an examination of the mucus that is coughed up or taken from the lungs to check for the presence of cancerous cells and blood tests allow for the identification of markers that are correlated with lung cancer.

Therapy

The most common type of treatment is surgery. During surgery, a surgeon works to remove lung cancer and a margin of healthy tissue. Surgery is often recommended only if the cancer is confined to small parts of the lungs. The most common types of surgery are wedge resection, segmental resection, lobectomy, and pneumonectomy.

Wedge Resection	A small section of the lungs containing the targeted tumor is removed along with a margin of healthy tissue.
Segmental Resection	A portion larger than the one in wedge resection but smaller than an entire lobe is removed.
Lobectomy	One out of the 5 lobes of the lungs is removed.
Pneumonectomy	An entire lung is removed

Another form of therapy is radiation therapy. Radiation therapy uses radiation such as high-powered energy beams from sources such as X-rays and protons to kill cancer cells. It is often used to shrink the target tumor before surgery or to kill any cancerous cells left in the lungs after surgery. If a patient is not healthy enough for chemotherapy and if the cancer is spread outside the lungs, radiation therapy is often recommended as the primary treatment.

Chemotherapy uses anti-cancer drugs to kill cancer cells. One or a combination of two or more drugs can be given through a vein in the arm through IV infusion, or given orally. Usually, a combination of drugs is given over some time with breaks between each treatment to give the patient time to recover. It's often used after surgery, along with radiation therapy, to kill any cancer cells that may be left. It's

sometimes also used before surgery to shrink the size of the tumor and make it easier to move it during surgery.

Stereotactic Body Radiotherapy is an intense version of radiotherapy that involves aiming many beams toward the cancer from different angles. It's usually an option for people who are not strong and healthy enough to go through surgery. Furthermore, stereotactic body radiotherapy can be used when lung cancer has spread to other body parts such as the brain.

Small molecule therapies are another option. Epidermal Growth Factor Receptor (EGFR) are proteins found on the surface of a cell or the inside of a cell. Normally, the Epidermal Growth Factor attaches to the EGFRs inside the cell and sends signals to the tyrosine kinase to allow the cell to grow and divide. However, the EGFR such as Tyrosine Kinase Inhibitors stop this signaling, which prevents the cancerous cells from growing and dividing. Some Tyrosine Kinase Inhibitors being used/researched by labs such as Yale's Politi lab to treat lung cancer include Dabrafenib, Trametinib, Crizotinib, and more.

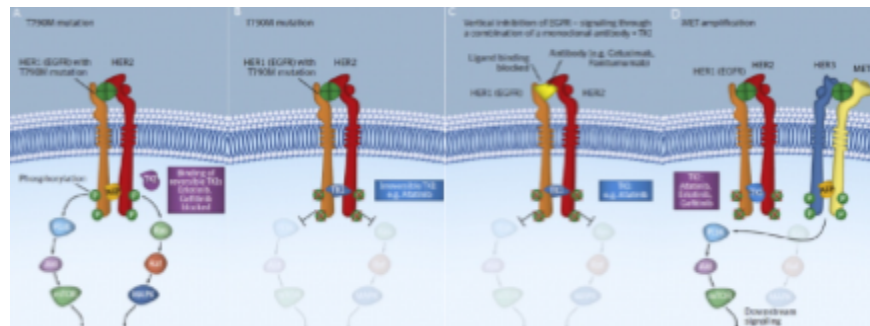


Figure 4: Visual of how the EGFR inhibitors work in preventing cancerous cells from growing.

Contrary to small molecule therapies, there are also large molecule therapies. Lab-produced immune system proteins called monoclonal antibodies are used to treat a variety of illnesses, including cancer. These antibodies bind to cell growth-promoting targets during cancer therapy. These antibodies stop malignant cells from proliferating or spreading. Treatments that block the formation of blood vessels, such as VEGF inhibitors, can kill certain cancer cells that depend on blood vessels

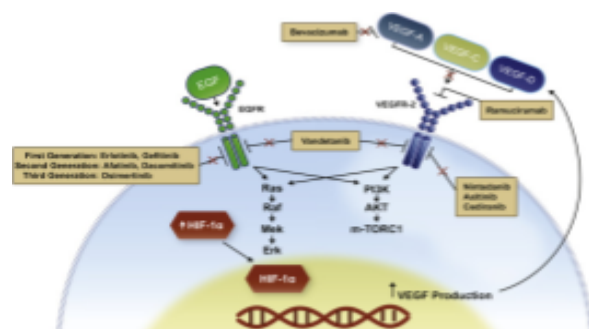


Figure 5: Demonstrates how the monoclonal antibodies are used to inhibit VEGF-A proteins.

for survival. Both the EGFR and VEGFR-2 signaling pathways can initiate the RAS/RAF/ERK and PI3K/AKT pathways. Therapeutic agents are being developed to inhibit the VEGF and EGFR pathways (indicated by brown boxes). These agents include monoclonal antibodies that either inhibit the VEGF-A protein (bevacizumab) or interfere with the extracellular domain of VEGFR-2 (ramucirumab).

Suggestions

NSCLC stands as a complex and heterogeneous disease that requires unique approaches to the different diagnoses and treatments for each patient's case. While treatment and therapies like targeted therapies and Immunotherapies have been well developed and help to lower the effects of progression development of cancer over time, not all treatments/ therapies have definite predictive outcomes among each patient. There stands a need for proper indication of the right treatment and therapies that can best be used to treat each patient, while also allowing to assess the effectiveness of each treatment.

Critical Need 1

Certain tests have shown that mutations and other abnormalities, such as the Epidermal Growth Factor Receptor (EGFR) mutation, among certain NSCLC patients, lead to resistance to some of the targeted therapies used to treat the cancer. With the uncontrolled growth of cancerous cells, the mutation added to the growth factor and led to an increased rate of metastasizing of cells to other body parts. Using chemotherapy and other treatments showed little progress in managing their progression-free lung cancer.

Critical Need 2

In NSCLC, mutations within the epidermal growth factor receptor (EGFR) gene present a critical point of concern. Despite initial efficacy, the development of resistance to osimertinib poses a significant clinical challenge. Mechanisms of resistance, including alterations in the MET gene and EGFR pathways, underscore the need for innovative therapeutic strategies beyond standard treatment protocols.

Solution

For the first critical need, the advancement in research on biomarker tests allowed for the mutation specifics and mechanisms to be discovered and for targeted antibody-drug therapy, to be used specifically for patients with this mutation. In a lab test conducted and published in the *Annals of Oncology*, data showed that “among the patients who were discovered to be EGFR mutants, those who were given the antibody along with regular chemotherapy treatment showcased progress-free cancer survival, compared to those who were just given chemotherapy, due to their resistance” (*Annals of Oncology* 77-90). With the help of the biomarker test, the mutation was discovered, and therapy research using antibodies that best act against the resistance in each patient.

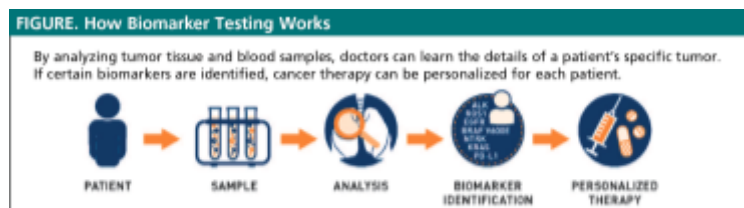


Fig 6: Garon E., Ledezma B. Biomarker Testing in Non-Small-Cell Lung Cancer. *Conquer Vol 6*, pg. 1-3 (2020).

Showcases the overview of the process of Biomarker testing and in direction toward personalized therapy.

For the second critical need, amivantamab, an EGFR-MET bispecific antibody, and lazertinib, a highly selective third-generation TKI, offer promising solutions for addressing the limitations of current treatment paradigms in EGFR-mutated NSCLC. Amivantamab, with its multiple mechanisms of action including ligand blocking, receptor degradation, and engagement of effector cells, demonstrates versatility in targeting activating and resistance mutations in EGFR-mutated NSCLC. Its bispecific nature allows for simultaneous targeting of MET, a common mechanism of resistance. Lazertinib, on the other hand, targets activating EGFR mutations and T790M resistance, offering complementary mechanisms to amivantamab and potentially enhancing therapeutic outcomes.

The paper states that “combining amivantamab with lazertinib has shown clinically meaningful activity in patients with EGFR-mutated NSCLC after disease progression on osimertinib. . . The addition of carboplatin–pemetrexed (chemotherapy) to amivantamab or amivantamab plus lazertinib represents a multifaceted approach to counter osimertinib-based resistance (*Annals of Oncology*).” Early-phase studies

have demonstrated promising objective response rates, laying the groundwork for further evaluation and validation in larger-scale trials such as MARIPOSA-2. Through rigorous assessment of the efficacy and safety of amivantamab–chemotherapy and amivantamab–lazertinib–chemotherapy regimens, MARIPOSA-2 aims to redefine treatment standards for patients with EGFR-mutated advanced NSCLC facing limited therapeutic options post-osimertinib monotherapy.

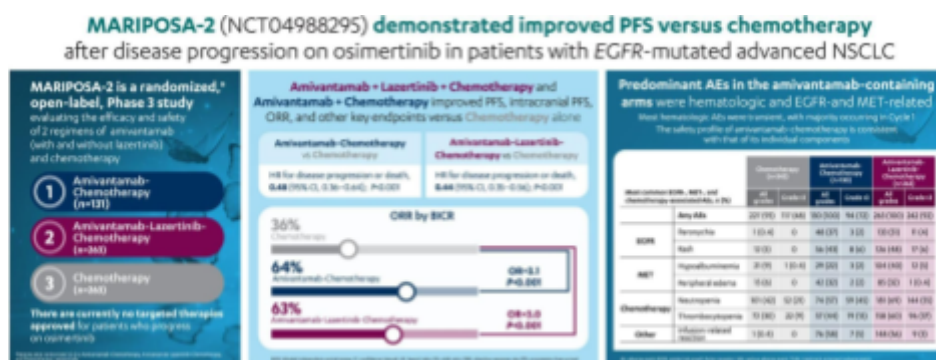


Figure 7: Data of Clinical Trials

While not a definitive solution, the integration of amivantamab and lazertinib represents a pivotal stride towards addressing the unmet clinical needs in EGFR-mutated NSCLC, offering renewed hope and therapeutic avenues for patients navigating the complexities of advanced lung cancer management.

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